Arunabha Majumder

Engineer | Experimental Data Acquisition, Sensor Data, Python and MATLAB

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Application: Data Engineer

Dear Hiring Team,

I am applying for the Data Engineer position. I am finishing my PhD in Mechatronics and Robotics at Aalborg University, and I hold an MTech in Mechatronics and a BTech in Mechanical Engineering. My PhD ends in March.

A big part of my PhD has been getting good data out of real experiments. For my shoulder exoskeleton study, I used the Qualisys data acquisition system to record EMG from 12 participants across different assistance modes and loads. In a second study, I used Qualisys motion-capture cameras to track arm motion while people interacted with a 5-bar planar parallel robot, so we could quantify human arm impedance with 6 participants. To line up the two data streams, I built an Arduino-based trigger that synchronised the robot's motion with the Qualisys recording.

That work taught me what data engineering looks like in practice. EMG, motion capture and robot sensor data all come in at different rates, from different systems, and they only make sense if they are recorded, labelled and stored carefully for every session. I set up the recordings, kept the data organised per participant and condition, and made sure it was ready for the people doing the analysis. I work in Python and MATLAB.

I want to be honest that my background is experimental and hardware-focused. I have not yet built production data pipelines with tools like SQL databases or cloud platforms, and I would like to learn them in a team that works with real sensor data.

Thank you for considering my application. I would be glad to talk about how my experience with sensor data could help your team.

Kind regards,
Arunabha Majumder